Obstacle avoidance car

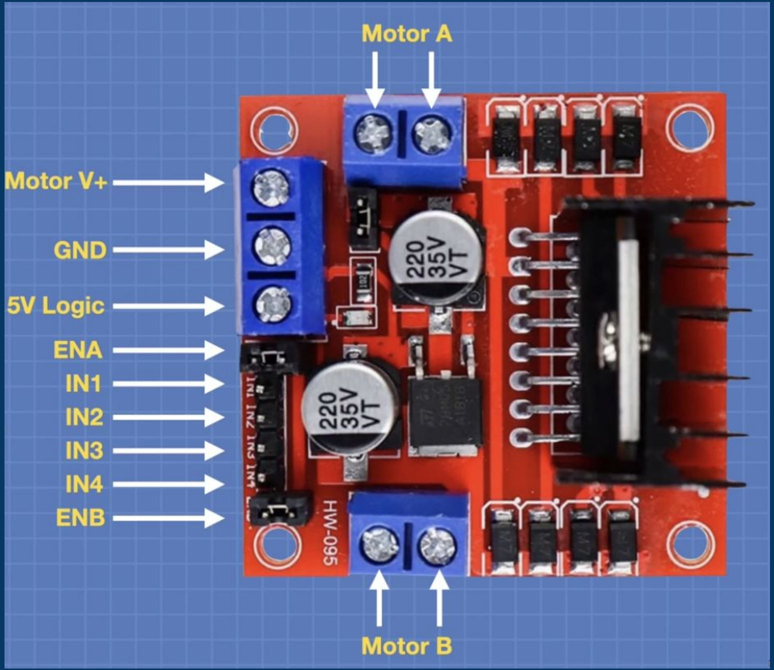
Motor Driver - L298N

Power

- V+ is motor voltage
- 5v Logic needs external power or Jumper for internal 5v regulator from Motor power

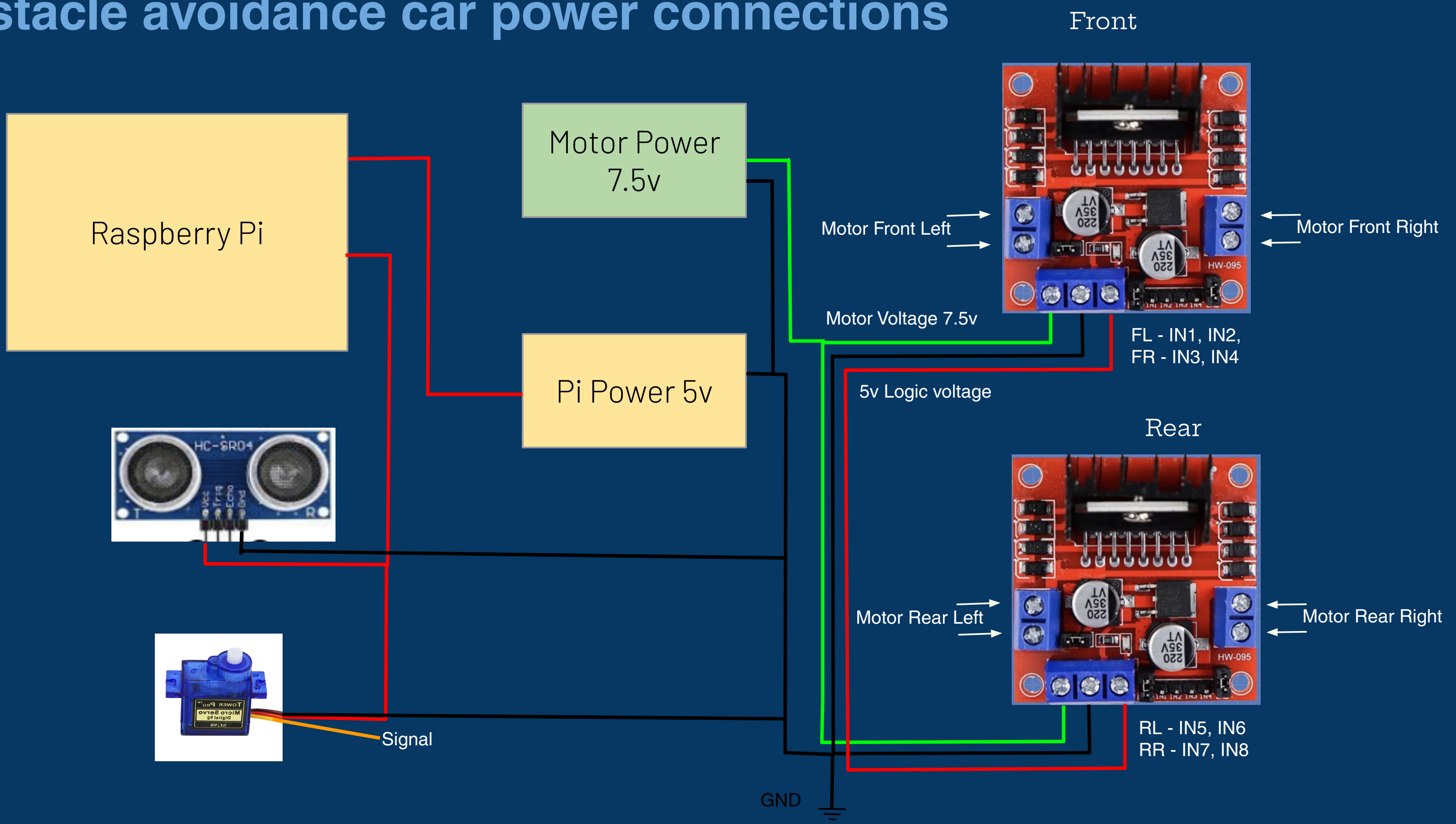
Motor connections

- Motor A is mapped to IN1and IN2
- Motor B is mapped to IN3 and IN4
- ENA is for speed control of A, ENB:B
- Single Speed - PWM jumpers are on (not used)



IN1 (or IN3)	IN2 (or IN4)	Direction
LOW	LOW	OFF
HIGH	LOW	Forward
LOW	HIGH	Reverse
HIGH	HIGH	Brake (OFF)

Obstacle avoidance car power connections



Obstacle avoidance car - logic connections (Pin map)

Wheel	Pin A	PIN B
Back-Right (RR - IN7, IN8)	GPIO 17	GPIO 27
Back-Left (RL - IN5, IN6)	GPIO 5	GPIO 6
Front-Right (FR - IN3, IN4)	GPIO 13	GPIO 19
Front-Left (FL - IN1, IN2)	GPIO 20	GPIO 21

Device	Pin	Direction
HC-SR04 TRIG	GPIO 23	Pi -> sensor (output)
HC-SR04 ECHO	GPIO 24	sensor -> Pi (input, level shift to 3.3v)
Pan servo (signal)	GPIO 18	Pi -> servo (PWM, 50Hz)
Buzzer	GPIO 22	Pi -> buzzer (PWM, active-tone)